



Low-density spruce plantations increase foraging by moose in a northeastern temperate forest



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ABSTRACT

Several North American jurisdictions are adopting new forest management practices that favor sustainable development and biodiversity conservation, but that are also expected to reduce timber yield. In many regions, forest managers have identified areas of intensive timber production where spruce plantations and thinning treatments are to be used extensively. Concomitantly, wildlife managers are concerned that intensive silvicultural treatments will have a negative impact on sport hunting by modifying the behavior and abundance of game species. We evaluated whether an experimental low-density spruce plantation (1350 stems/ha) could increase habitat suitability for moose *Alces americanus* compared to standard high-density plantations currently used throughout eastern Canada (~2000 stems/ha). We evaluated the effects of plantation type, pre-commercial thinning, and age class of plantations on proxies of habitat suitability and use by moose in a northeastern temperate forest of Bas-Saint-Laurent, Québec (Canada). We also considered various environmental characteristics measured locally and within 3.14-km² landscape contexts to explain spatial variations in use by moose among 540 sampling plots. Our results show that the proportion of stems browsed by moose increased with the number of available stems and decreased with vertical cover. The number of moose feces, an index of time spent by moose in sampling plots, was similar in both plantation types, but was lower in thinned and young (5 years old) stands as compared to unthinned and 15 years old stands. Variations in the proportion of browsed stems and the number of feces were not explained by broad-scale indices of food availability, density of edges between stands offering good cover and foraging opportunities, road density, elevation, or the area occupied by plantations and naturally regenerated stands around sampling plots. Moose used low-density plantations significantly more than standard plantations for browsing, and these stands supported higher stem densities and vertical cover than standard high-density plantations on average. Low-density plantations offered good foraging opportunities for moose by allowing a large variety of palatable deciduous species to grow between planted coniferous stems. However, we found high inter-site variability in environmental characteristics following plantation, even among equivalent treatments. This suggests that forest managers should aim for particular stand characteristics (i.e. high availability of stems and low vertical cover) *in situ* rather than prescribing specific post-harvest silvicultural treatments (e.g., plantation types) to favor moose. We recommend the use of unthinned low-density plantations in northern temperate regions where the management objective is to maintain high moose hunting success.

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1. Introduction

During the 20th century, the goal of most North American forest services has been to maximize annual allowable timber yield. In many jurisdictions, including Québec, Canada, this practice has persisted throughout the beginning of the 21st century, leading to a relative homogenization of forest structure and composition, particularly in intensively managed areas (Coates and Burton,

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1997; Schulte et al., 2007). More recently, ecosystem-based management has been suggested as a way to improve resource extraction practices, including logging (Slocumbe, 1993; Gauthier et al., 2008). This approach seeks to reach equilibrium between socioeconomic development and biodiversity conservation by emulating natural disturbances and broad-scale changes that prevailed during the precolonial era. As such, ecosystem-based management is expected to reduce allowable timber yield, which has led the industry to express concerns. In some regions, intensive timber production areas (hereafter ITPA) have been suggested as a way to compensate this potential economic loss. In ITPA, forest managers aim to use conifer plantations and repeated thinning operations to allow faster harvest rotations and higher timber yield (Lieffers et al., 2003). Although post-harvest silviculture has been shown to enhance wood supply (Pelletier and Pitt, 2008) and stem quality (Mäkinen and Isomäki, 2004), several studies have highlighted the negative impacts of intensive forest management on biodiversity (e.g., see reviews by Thompson et al., 2003 and Carnus et al., 2006). Notably, declines in habitat suitability have been reported for animals living in exotic tree plantations and monocultures, including forest birds (Hayes et al., 2003), small mammals (Homyack et al., 2005), and ungulates (Gill et al., 1996).

Depredation of plantations by wild herbivores is well documented, mainly because of its negative economic impacts (e.g., Heikkilä and Härkönen, 1996; McLaren et al., 2000). However, few studies have investigated the importance of conifer plantations in the habitat selection of wild ungulates. Gysel (1966) found that populations of white-tailed deer (*Odocoileus virginianus*) declined in areas planted with exotic red pines (*Pinus resinosa*) in Michigan. Staines and Welch (1984) found that red deer (*Cervus elaphus*) in Scotland preferred newly established Sitka spruce (*Picea sitchensis*) plantations over their traditional open-hill habitat. Peek et al. (1976) observed no particular selection of conifer plantations by moose (*Alces americanus*) year-round in northeastern Minnesota, although low-density plantations (<124 stems/ha) were selected during summer. According to these authors, reforestation became increasingly negative for moose populations as conifer stocking rates increased and available forage (i.e., low-ground regenerating trees and shrubs) decreased. More recently, Leclerc et al. (2012) demonstrated that moose densities in southeastern Québec increased with the proportion of spruce plantations in 60-km² aerial survey sampling plots. They also found fewer moose in areas where plantations were aggregated, suggesting that landscape context was an important driver determining the use of plantations by wild ungulates.

Moose is an important wildlife species across northern North America because of its ecological and economic roles. In eastern Québec, mainly in the Bas-Saint-Laurent (hereafter BSL) region, moose have reached unprecedented numbers. The region has a long and diversified history of forest management (>200 years; Boucher et al., 2009), which culminated with severe eastern spruce budworm (*Choristoneura fumiferana*) epidemics and concurrent salvage logging activities during the mid-1970s to mid-1990s (Boulanger and Arseneault, 2004). These large scale timber harvest operations have increased the representativeness of regenerating deciduous stands, which provide excellent forage and cover for moose (Potvin et al., 2005). Population growth has also been favored by the absence of gray wolf (*Canis lupus*) – the main predator of adult moose – since the late 1800s. Some particularly suitable areas now show densities of up to 33 moose/10 km², which are comparable to some of the highest densities in the country (Franzmann and Schwartz, 2007). Moose sport hunting has become a rapidly growing industry in BSL (benefits of ~26 M\$ in 2007; CRE-BSL, 2010), with hunters achieving excellent success rates each year. In contrast, the timber economy (benefits of ~83 M\$ in 2005; CRE-BSL, 2010) has slowed drastically during the last

decade due to a generalized “forest crisis” originating from concerns about the sustainability of the province’s forestry practices, declining demands from the U.S. market, and constraining laws and regulations (Barré and Rioux, 2012). From the perspective of wildlife managers, hunters, and outfitters, ITPA are seen as detrimental to the moose and its hunt. Consequently, there is a real need to assess the effectiveness of new management practices that could favor timber harvesting while maintaining relatively high moose densities and hunting success in ITPA.

Our objective was to evaluate if, and to what extent, an experimental low-density spruce plantation could increase habitat suitability for moose as compared to standard high-density plantations currently used throughout eastern Canada. High-density plantations are generally used as intensive silvicultural practices to maximize wood supply and profitability (Lieffers et al., 2003), but a pioneer study by Peek et al. (1976) suggested that low-density plantations could show characteristics favorable to moose. Plantations are often followed by thinning treatments to maximize the survival probability of planted trees against competing regeneration (Mäkinen and Isomäki, 2004). We therefore investigated the effect of pre-commercial thinning on habitat suitability and use by moose in both types of plantations (i.e., low- and high-density), as well as naturally regenerated stands. Finally, for each plantation type and thinning treatment, we sampled several replicates of varying age classes to document potential changes in use by moose through time. We performed analyses at fine and broad scales to consider most factors known to influence space use by moose, including forage and cover availability (Bjørneraas et al., 2012), interspersed between forage and cover (Dussault et al., 2006), topography (Leblond et al., 2010), and roads (Beyer et al., 2013).

2. Materials and methods

2.1. Study area

Our study area (2300 km²) was located in the Bas-Saint-Laurent (BSL) region in Québec, Canada, southeast of Rimouski (Fig. 1). The area lies in the balsam fir (*Abies balsamea*) – yellow birch (*Betula alleghaniensis*) bioclimatic domain of the northern temperate forest, where yellow birch, white birch (*B. papyrifera*), balsam fir, white spruce (*Picea glauca*), white cedar (*Thuja occidentalis*), and sugar maple (*Acer saccharum*) are the main tree species. The understory is also composed of mountain maple (*A. spicatum*), hobblebush (*Viburnum lantanoides*), beaked hazel (*Corylus cornuta*), red elderberry (*Sambucus racemosa*), and Canadian yew (*Taxus canadensis*). Black (*P. mariana*) and white spruce plantations make up 13% of public forests in BSL (Leclerc et al., 2012), but they could represent more than 25% of the land following the establishment of ITPA in the region. Based on an aerial survey conducted in February 2014, moose density in BSL hunting zones (i.e., at the regional scale) was estimated at 11.4 ± 1.7 individuals/10 km², with local peaks reaching 33 moose/10 km² in particularly favorable areas (Ross et al., 2014). However, moose density was relatively high and homogeneous throughout our study area. Black bear (*Ursus americanus*) and coyote (*Canis latrans*) are potential predators of moose calves, but gray wolf is absent from the region.

2.2. Plantation types

The experimental low-density plantation at 1350 stems/ha (hereafter P1350) was originally designed to prevent the transition from mixed to conifer-dominated stands in the northern temperate zone (Laflièche et al., 2000). This approach, developed in 1995 by the BSL Model Forest (previously part of the Canadian Model Forest Network), advocates the use of a lower tree density to

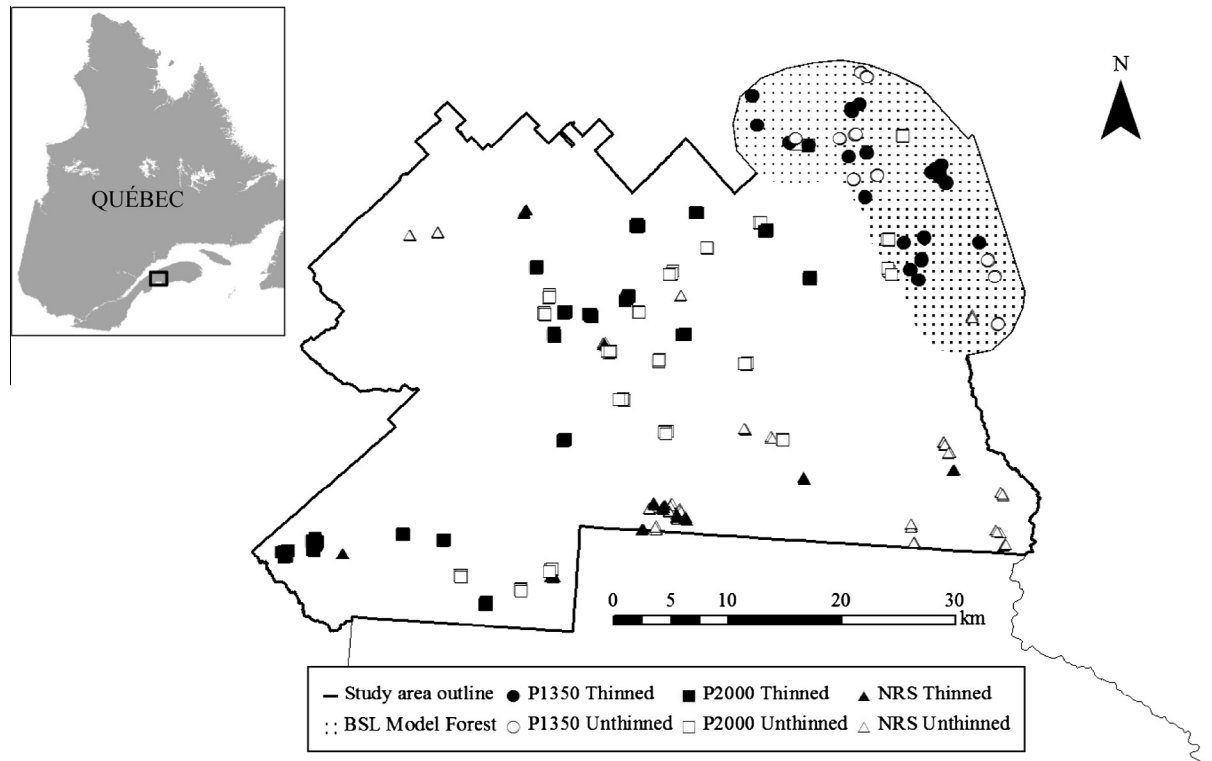


Fig. 1. Study area where we evaluated the effect of plantation type, thinning treatment, and age class on proxies of habitat suitability for moose, use by moose, and forest productivity in Bas-Saint-Laurent, Québec, Canada. We randomly drew 540 sampling plots in 108 sites representative of the possible combinations of 3 plantation types (experimental low-density: 1350 stems/ha [P1350], standard high-density: 2000 stems/ha [P2000], or naturally regenerated stands [NRS]) $\times$ 2 thinning treatments (thinned or unthinned) $\times$ 3 age classes (5, 10, or 15 years old; not represented here). Sample sizes for each combination are presented in [Appendix A](#).

increase habitat suitability and connectivity for a large variety of animal species (Belleau and Tremblay, 2003). In the P1350, trees are planted 2 m apart in rows set 3.7 m apart (see [Fig. 2](#)). Rows are cleared (1 m on each side) of any competing regeneration 2–3 years after the plantation, leaving a 1.7-m band of natural regeneration, usually deciduous, between rows of planted spruce trees. The P1350 also uses large white spruce seedlings (>35 cm in height) to increase their probability of surviving against competing natural regeneration. These larger seedlings exhibit a higher growth rate than standard seedlings, which are usually 20–22 cm tall (Thiffault, 2004). The P1350 was only used in mixed stands of the BSL Model Forest, situated in the northeastern part of our study area (see [Fig. 1](#)).

The standard high-density plantation at 1875–2200 stems/ha (2000 stems/ha on average; hereafter P2000) is commonly used to rapidly re-establish a coniferous forest cover following harvest. In this plantation type, standard seedlings are planted every 2 m in all directions to maximize stocking (see [Fig. 2](#)). The P2000 was used in conifer-dominated stands all across the study area. We compared the vegetation structure of both types of plantations to that of naturally regenerated stands (hereafter NRS), which we used as controls. NRS were harvested using the same technique (i.e., cutting with protection of regeneration and soils, hereafter CPRS) and during the same period (last 6–18 years), but were not followed by a plantation.

2.3. Field surveys

We conducted field surveys from May 14 to July 31, 2013. We aimed at sampling an equal number of sites among the 18 possible combinations of 3 plantation types (P1350, P2000, and NRS) $\times$ 2

thinning treatments (thinned and unthinned stands) $\times$ 3 age classes. Potential sites were plantations ($n = 2957$) or naturally regenerated stands ($n = 630$) located in 6–18 years old CPRS. We grouped plantations made in 1995–1997, 2000–2002, and 2004–2007 in 3 age classes of approximately 15, 10, and 5 years old, respectively. We randomly selected 7 sites per plantation $\times$ thinning $\times$ age combination, in which we placed 5 sampling plots at random locations. Due to the rarity of some combinations (notably the absence of thinned 5 years old NRS), we sampled a total of 108 sites and 540 plots (see [Appendix A](#) and [Fig. 1](#)).

We counted the number of unbrowsed and recently browsed stems within a 16 m² circular subplot (2.26-m radius) centered on each of the 540 sampling plots. We counted all >10 cm tree twigs located between 30 cm and 3 m from the ground to consider the average reach of a feeding moose. We counted the number of fresh (≤ 1 year) moose feces (number of piles) on a 400-m² circular plot (11.28-m radius) centered on each sampling plot (St-Louis et al., 2000). We used the proportion of browsed stems and the number of feces as indices of foraging activity and time spent by moose in sampling plots, respectively. Both measures were indicative of winter and early spring use, i.e., periods when moose browse on woody twigs and leave feces in the form of small nuggets (Franzmann and Schwartz, 2007). We did not count old feces (>1 year), which were easily identifiable by their color and location in the litter (i.e., old feces are usually covered by litter). Within the same 400-m² plot, we also measured variables of stand composition and structure which we used as proxies of habitat suitability for moose. An observer located at the center of the plot estimated lateral cover using a 2-m vegetation profile board held at 15 m in four cardinal directions (Nudds, 1977). The same observer estimated vertical cover through a 50 $\times$ 50 cm frame held 2 m

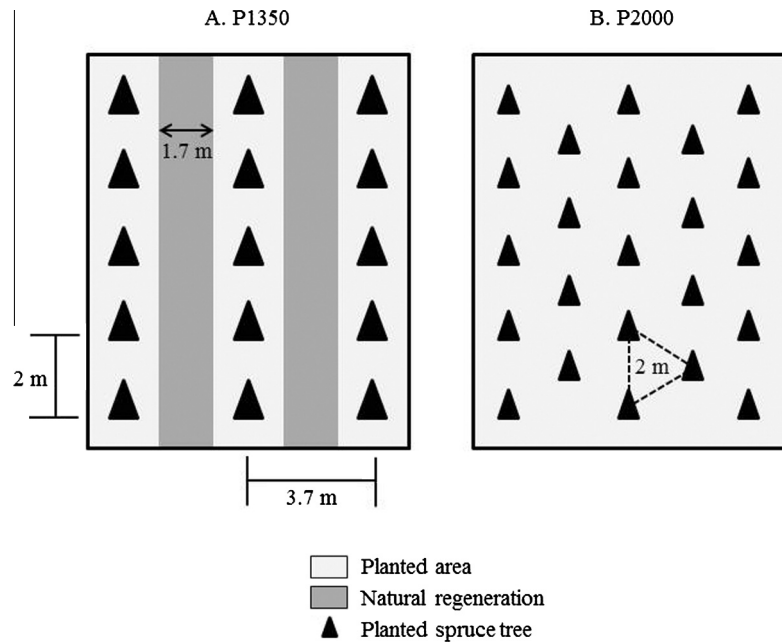


Fig. 2. Schematic representation of the two plantation types evaluated in this study. A: The experimental low-density plantation (P1350) uses large spruce seedlings (>35 cm in height) spaced 2 m apart and planted in rows 3.7 m apart, for a density of 1350 stems/ha on average. B: The standard high-density plantation (P2000) uses standard seedlings (20–22 cm in height) planted every 2 m in all directions to maximize coniferous stocking, for a density of 1875–2200 stems/ha (2000 stems/ha on average).

above the ground in the center of the plot as well as 15 m from the center in the four cardinal directions (St-Laurent et al., 2008). Lateral and vertical covers were estimated in 10-% classes, from no obstruction (0%) to complete obstruction (100%). We counted and identified each tree and calculated the density of commercial trees (no./ha). Finally, we measured the diameter at breast height (DBH, cm) of all commercial trees, and measured the height (m) of 5 codominant trees.

2.4. Landscape context

We generated circular buffers of 1-km radius around each site (hereafter landscape contexts) using a geographic information system (ArcGIS 10, ESRI Inc., Redlands, CA). We define landscape context as the habitat immediately available to or susceptible to influence the behavior of animals. We used digital maps provided by the MFFP to determine the proportion of different land cover types and assess various environmental characteristics within each of these landscape contexts. These maps were derived from aerial photographs taken in 2004 at the scale of 1:20000, with a minimum mapping unit of 4 ha for forest stands and 2 ha for non-forested areas (e.g., lakes, unproductive lands). We combined vegetation classes into 11 stand types using dominant cover type and age class of forest stands, based on a classification by Dussault et al. (2006; see Appendix B). We used the food availability index (hereafter food index) developed by Dussault et al. (2006; p. 1099, Eq. 1) to estimate the proportion of stands providing high food availability (Mi10, Mt50, and Dt50) within each landscape context. We calculated the density of edges (km/km²) between stands offering good cover (Mi50, CF30, and CS30) and good foraging opportunities (see Appendix B). Based on Leclerc et al. (2012), we calculated the proportion of plantations (P1350 and P2000 taken together) and naturally regenerated stands in each landscape context, as well as road density (km/km²) and mean elevation (m) obtained from a digital elevation model. Finally, we calculated principal coordinates of neighbor matrices (PCNM) to account for autocorrelation among sampling plots (Borcard and Legendre, 2002). We used the *x* and *y* coordinates of the plot centroids to generate a

truncated distance matrix (see Borcard and Legendre, 2002), and modeled the spatial structure of our dataset by generating as many PCNM eigenvectors as there were potential spatial scales in our experimental design. We then included all statistically significant PCNM eigenvectors as independent variables in our models (St-Laurent et al., 2008).

2.5. Statistical analyses

We compared the relative influence of various silvicultural, environmental, and anthropogenic variables on the proportion of browsed stems and the number of moose feces in sampling plots, which we used as proxies of habitat use. We used Akaike's information criterion corrected for small sample sizes (AIC_c; Burnham and Anderson, 2002) to hierarchically evaluate the relative empirical support received by 14 candidate linear regression models composed of various local- (i.e., measured *in situ*) and landscape-scale (i.e., measured in 3.14-km² landscape contexts) covariates (Table 1). We used a first set of models to determine the influence of variables measured within the 3.14-km² landscape contexts around sampling plots. Landscape-scale variables were the proportion of stands providing high food availability, density of edges between food and cover stands, density of roads, mean elevation, and area occupied by plantations and naturally regenerated stands. We kept the most parsimonious model from this first set of models (i.e., *landscape context* models, see Table 1) and compared it to other models composed of variables measured on a local scale. The variables represented (1) plantation type, thinning treatment, and age class (*silvicultural treatments* model), and (2) field measurements indicative of habitat suitability for moose (i.e., number of stems, lateral cover, and vertical cover; *field measurements* model). We also considered all possible combinations of these models as well as a null model (see Table 1). All models included statistically significant PCNM eigenvectors as fixed effects and the site as a random effect. We performed collinearity diagnostics and found that the variance inflation factor was low (≤ 1.81) in all models.

Table 1

Candidate models used to evaluate the effect of local and landscape-scale variables on the proportion of browsed stems and the number of moose feces in sampling plots in Bas-Saint-Laurent, Québec, Canada. All models included 13 statistically significant PCNM eigenvectors to account for spatial autocorrelation between sample sites. k = number of parameters, excluding PCNM eigenvectors.

Model	Variables	k
<i>Landscape context</i>		
LAND1	Food index + density of edges (km/km ²)	2
LAND2	Density of roads (km/km ²)	1
LAND3	Mean elevation (m)	1
LAND4	Proportion of the area occupied by plantations (P1350 + P2000;%) + proportion of the area occupied by naturally regenerated stands (NRS;%)	2
LAND5	Model LAND1 + model LAND2	3
LAND6	Model LAND1 + model LAND2 + model LAND3	4
LAND7	Model LAND1 + model LAND2 + model LAND3 + model LAND4	6
<i>Silvicultural treatments</i>		
TREAT	Plantation type (P1350, P2000, NRS) + thinning treatment (thinned or unthinned) + age class (5, 10, or 15 years old)	3
<i>Field measurements</i>		
FIELD ^a	Number of stems + lateral cover (%) + vertical cover (%)	3
<i>Model combinations</i>		
LAND + TREAT	Landscape context ^b + Silvicultural treatments	4
LAND + FIELD	Landscape context ^b + Field measurements	4
TREAT + FIELD	Silvicultural treatments + Field measurements	6
LAND + TREAT + FIELD	Landscape context ^b + Silvicultural treatments + Field measurements	7
<i>Null model</i>		
NULL	PCNM eigenvectors only	0

^a All variables from the *Field measurements* model were log-transformed.

^b Only the most parsimonious model from the *Landscape context* set was used (model LAND2; see Table 2).

To better inform management decisions, we assessed whether discrete silvicultural treatments influenced indices of habitat suitability for moose (i.e., number of stems, lateral cover, and vertical cover), use by moose (i.e., proportion of browsed stems and number of feces), and forest productivity (i.e., density, DBH, and height of commercial trees). We used this *a posteriori* analysis to investigate variations among sites within and between treatments, i.e., to highlight potential differences between “desired” and observed characteristics of treated sites. We fitted generalized linear models using plantation type, pre-commercial thinning, stand age, and all possible double interactions as independent variables. We log-transformed all dependent variables and used the site as a random effect in all analyses. We compared least squares means with Satterthwaite’s approximation for degrees of freedom to investigate the differences between levels of factors. We conducted all statistical analyses using R 3.0.2 (R Core Team, 2013).

3. Results

3.1. Modeling the influence of silvicultural, environmental, and anthropogenic variables using information theory

Our spatial analysis isolated 273 PCNM eigenvectors, 13 of which were statistically significant and included into all candidate models. The most parsimonious landscape context model explaining variations in the proportion of browsed stems and the number of moose feces among sampling plots was a model composed of road density only (i.e., model LAND2 had the lowest intra-group ΔAIC_c value for both dependent variables, see Table 2). When considering other sets of models, we found that variations in the proportion of browsed stems and the number of moose feces were

best explained by models composed of field measurements and silvicultural treatments, respectively (see the lowest inter-group ΔAIC_c values in Table 2).

Moose browsed a greater proportion of stems in stands where the number of available stems was high and where the vertical cover was low (Table 3). Fewer feces were found in plantations as compared to NRS, in thinned stands as compared to unthinned stands, and in 5 years old stands as compared to 15 years old stands (Table 3). We found high inter-site variability in the availability of stems and vertical cover (i.e., the two covariates that explained most of the variation in the proportion of browsed stems) among our various plantation $\times$ thinning $\times$ age class combinations (Fig. 3). Because of this variability among treated sites, we could not easily identify a single “best” silvicultural treatment for moose.

3.2. Differences among plantation types, thinning treatments, and age classes

Our *a posteriori* comparisons revealed that the P1350 had more stems [37.96 ± 2.09 (mean $\pm$ standard error, SE) vs. 29.22 ± 1.52 stems; $t = 2.57$, $df = 94$, $P = 0.012$] and a denser vertical cover (36.03 ± 2.31 vs. $19.17 \pm 1.42\%$; $t = 3.20$, $df = 94$, $P = 0.002$) compared to the P2000. The P1350 ($0.44 \pm 0.03\%$) also showed a higher proportion of browsed stems compared to the P2000 ($0.31 \pm 0.02\%$; $t = 3.20$, $df = 94$, $P = 0.002$) and NRS ($0.33 \pm 0.02\%$; $t = 2.08$, $df = 94$, $P = 0.040$). As outlined previously, the number of moose feces did

Table 2

Most parsimonious models evaluating the effect of local and landscape-scale variables on the proportion of browsed stems and the number of moose feces in sampling plots in Bas-Saint-Laurent, Québec, Canada. We present the log-likelihood (LL) and the difference between the Akaike’s information criterion corrected for small sample sizes value of the model and that of the most parsimonious model (ΔAIC_c). The most parsimonious model of the *Landscape context* set is shown in bold in the intra-group ΔAIC_c column, whereas the most parsimonious model overall is shown in bold in the inter-group ΔAIC_c column. The most parsimonious *Landscape context* model was used as a candidate model in inter-group comparisons. Variables composing each model are shown at Table 1.

Model	LL	Intra-group ΔAIC_c	Inter-group ΔAIC_c
<i>Proportion of browsed stems</i>			
LAND1	589.84	11.58	60.30
LAND2	594.56	0.00	
LAND3	590.93	7.27	
LAND4	586.49	18.28	
LAND5	585.49	21.71	
LAND6	577.88	39.82	
LAND7	566.71	66.49	
TREAT	597.22		63.59
FIELD	626.86		0.00
LAND + TREAT	594.11		71.99
LAND + FIELD	622.68		10.52
TREAT + FIELD	620.94		22.69
LAND + TREAT + FIELD	618.20		30.38
NULL	598.59		50.11
<i>Number of moose feces</i>			
LAND1	−139.38	11.95	15.53
LAND2	−134.47	0.00	
LAND3	−137.93	6.92	
LAND4	−136.42	6.03	
LAND5	−142.25	19.85	
LAND6	−148.37	34.24	
LAND7	−152.11	46.05	
TREAT	−122.40		0.00
FIELD	−132.70		16.27
LAND + TREAT	−125.00		7.37
LAND + FIELD	−135.60		24.23
TREAT + FIELD	−120.99		3.73
LAND + TREAT + FIELD	−123.66		11.27
NULL	−131.58		7.62

Table 3

Parameter estimates (β) and 95% confidence limits (95% CL) of the most parsimonious models evaluating the effect of local and landscape-scale variables on the proportion of browsed stems and the number of moose feces in sampling plots in Bas-Saint-Laurent, Québec, Canada. Statistically significant effects are characterized by 95% confidence limits excluding 0 and are shown in bold. All models also included 13 statistically significant PCNM eigenvectors which are not represented here.

Covariate	β	95% CL
<i>Proportion of browsed stems</i>		
Number of stems	0.075	[0.056; 0.095]
Lateral cover	−0.020	[−0.098; 0.057]
Vertical cover	−0.031	[−0.044; −0.019]
<i>Number of moose feces</i>		
Plantation – P1350 ^a	−0.370	[−0.567; −0.173]
Plantation – P2000 ^a	−0.284	[−0.398; −0.171]
Thinned stands ^b	−0.114	[−0.209; −0.018]
Age – 5 years old ^c	−0.140	[−0.258; −0.024]
Age – 10 years old ^c	−0.060	[−0.169; 0.048]

^a The reference category was naturally regenerated stands (NRS).

^b The reference category was unthinned stands.

^c The reference category was 15 years old stands.

not differ between plantation types (1.54 ± 0.18 in P1350 vs. 2.13 ± 0.20 feces in P2000; $t = -1.37$, $df = 94$, $P = 0.175$) and was much higher in NRS (5.81 ± 0.60 feces; $P \leq 0.001$). The P2000 showed a higher density of commercial trees (1124.16 ± 21.50 vs. 655.78 ± 26.92 trees/ha; $t = -6.92$, $df = 94$, $P < 0.001$), but smaller (5.23 ± 0.08 vs. 6.90 ± 0.20 m; $t = 3.04$, $df = 93$, $P = 0.003$) and narrower (5.85 ± 0.04 vs. 7.28 ± 0.07 cm; $t = 2.03$, $df = 91$, $P = 0.046$) trees on average compared to the P1350.

NRS had fewer stems (25.10 ± 1.56 stems on average) than both plantations types ($P \leq 0.031$), and the vertical cover ($29.05 \pm 1.71\%$ on average) was comparable to that measured in P1350 ($t = 0.78$, $df = 94$; $P = 0.437$). The density of commercial trees in NRS (1158.53 ± 27.73 trees/ha) was comparable to densities found in P2000 ($t = 0.68$, $df = 94$; $P = 0.497$) but denser than in P1350 ($t = -5.24$, $df = 94$; $P < 0.001$). The DBH of commercial trees in NRS (5.30 ± 0.05 cm) was significantly smaller than in both plantation types ($P < 0.001$), but their height (6.31 ± 0.11 m) was comparable ($P \geq 0.099$).

Less feces were found in thinned stands (2.26 ± 0.34 feces) compared to unthinned ones (4.00 ± 0.29 feces; $t = 2.17$, $df = 94$, $P = 0.032$). Thinned and unthinned stands had comparable stem availability (30.09 ± 1.43 vs. 30.95 ± 1.44 stems; $t = 1.80$, $df = 94$, $P = 0.076$), but thinned stands showed higher densities of commercial trees (1029.27 ± 18.27 vs. 961.27 ± 29.71 trees/ha; $t = -3.47$, $df = 94$, $P < 0.001$) and larger trees (i.e., DBH; 6.39 ± 0.04 vs. 5.41 ± 0.04 cm; $t = -2.15$, $df = 88$, $P = 0.034$) compared to unthinned stands.

Older stands had a lower proportion of browsed stems (0.22 ± 0.02 in 15-years-old, 0.39 ± 0.02 in 10-years-old, and $0.50 \pm 0.02\%$ in 5-years-old; $P \leq 0.074$) and showed the densest vertical cover (43.37 ± 1.60 in 15-years-old, 24.93 ± 1.65 in 10-years-old, and $7.59 \pm 0.98\%$ in 5-years-old; $P < 0.001$).

4. Discussion

We used complementary statistical approaches to evaluate the effect of environmental characteristics measured in and around two types of plantations and naturally regenerated stands on proxies of habitat use by moose. The first approach explored the relative influence of silvicultural, environmental, and anthropogenic variables using information theory. The second approach evaluated the influence of plantation type, thinning, and age class from a forest manager's perspective. Interestingly, silvicultural treatments were better predictors of variations in the number of moose feces in sampling plots, whereas proxies of habitat suitability *in situ*

were better predictors of variations in the proportion of browsed stems. We controlled for the spatial distribution of sampling plots in our study area by using PCNM eigenvectors (Borcard and Legendre, 2002). These covariates were often significant, suggesting that environmental gradients explained some of the variations in the use of sampling plots by moose. This approach allowed us to focus on the evaluation of specific environmental and anthropogenic variables.

4.1. Proportion of browsed stems

We found that moose browsed a greater proportion of stems in stands where their availability was high. We also found that the availability of stems was greater in experimental low-density plantations. Therefore, low-density plantations offered good foraging opportunities for moose, and indeed, our *a posteriori* comparisons allowed us to demonstrate that moose used them significantly more than standard plantations and naturally regenerated stands for browsing. However, we found that moose browsed more stems in stands having low vertical cover, and low-density plantations showed denser vertical cover compared to standard plantations. Vertical cover protects moose from high ambient temperatures during summer (Dussault et al., 2004) and limits snow accumulation during winter (Teti, 2003). In BSL, winters are relatively mild on average, and snow generally does not impede moose movement significantly. This result could also have been caused by the relatively large variability in vertical cover among sites for a given silvicultural treatment.

4.2. Number of moose feces

The number of moose feces did not differ between the two plantation types, suggesting that moose spent as much time in experimental and standard plantations. Naturally regenerated stands offered fewer foraging opportunities, but moose seemed to use them extensively as shown by the high number of feces found in these stands. Contrary to the proportion of browsed stems, which represented an indirect measure of foraging activity, the number of moose feces in sampling plots was not necessarily related to food availability. Moose (and ruminants in general) spend more than half of their time ruminating and resting (Risenhoover, 1986). Moose in our study area may have alternated use of different stand types (including different plantation types) daily to trade off food acquisition with protection from predation and harsh weather conditions during activities other than feeding (Houston et al., 1993).

4.3. Effect of pre-commercial thinning

Pre-commercial thinning had a negative effect on the availability of deciduous stems and the number of moose feces in sampling plots. Thompson et al. (1989) found that moose in Newfoundland selected for thinned fir stands to feed on longer, more digestible twigs. However, moose do not eat spruce twigs (Kurtila et al., 2002), and thinning performed in spruce plantations may reduce foraging opportunities for moose by inhibiting deciduous regeneration. On the other hand, thinning increased the density and diameter of commercial trees at our sites, a result also reported by Pothier (2002). Thinning may also increase the availability of deciduous twigs through the apparition of suckers and stump shoots, but this emergent regeneration was not sufficient to reverse the general decline in stem availability we observed in thinned stands in our study area. Pre-commercial thinning in temperate forests could increase forest productivity to the detriment of habitat suitability for large herbivores, but additional data would be required to test this hypothesis.

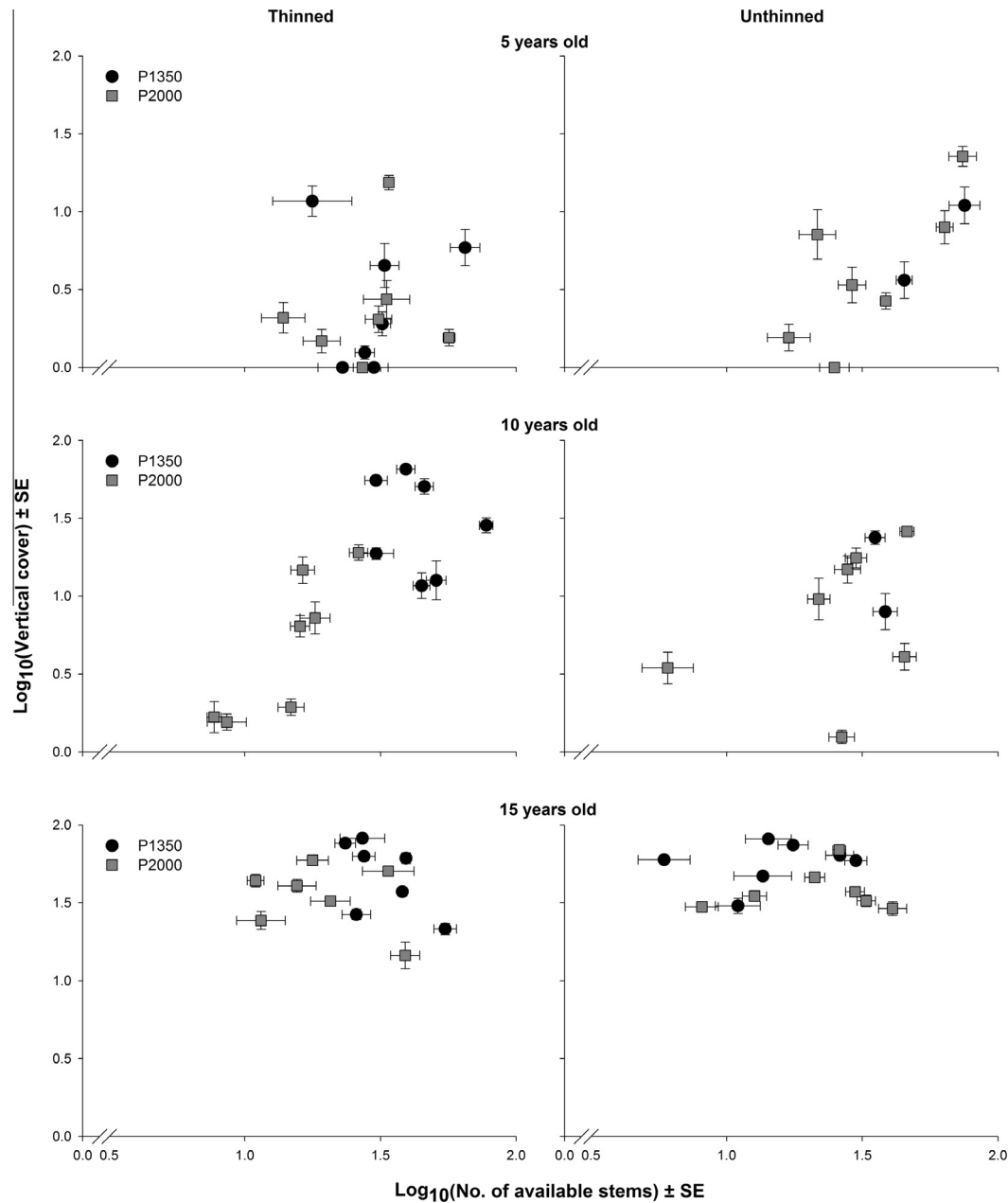


Fig. 3. Relationship between the number of available stems (log-transformed $\pm$ standard error, SE) and vertical cover (log-transformed $\pm$ SE) measured in sampling plots used to evaluate the effect of plantation type, pre-commercial thinning, and age class on proxies of habitat suitability and use by moose in Bas-Saint-Laurent, Québec, Canada. We compared experimental low-density plantations (1350 stems/ha [P1350]) to standard high-density plantations (2000 stems/ha [P2000]), and grouped plantations according to 2 thinning treatments (thinned or unthinned) and 3 age classes (5, 10, or 15 years old). Each dot (P1350 = black circle, P2000 = gray square) represents a sampling site.

4.4. Effect of landscape context

The use of sampling plots by moose was not explained by broad-scale variables. This could stem from the fact that our sample sites were not distributed across a wide range of landscape conditions, but rather within 18 specific combinations of plantation type $\times$ thinning treatment $\times$ age class. Moreover, experimental plantations were first used in 1995 (Belleau and Tremblay, 2003), limiting our range of age classes to the early stages of forest growth. Moose behavior could change over broader scales or as plantations get older, warranting the need to study these interactions at larger spatial scales and on the long term. For example, more insights could be gained from individual-based GPS telemetry monitoring of moose in areas of varying plantation densities (Cagnacci et al., 2010).

Food and the interspersed between food and cover were recognized by Dussault et al. (2006) as being the most important variables explaining habitat selection by moose in the balsam fir – white birch bioclimatic domain of the boreal forest. Moose were also shown to avoid areas of high road density in our study area (Leclerc et al., 2012) and elsewhere (Beyer et al., 2013). These variables were not good predictors of moose behavior in our study. We argue that food and interspersed between food and cover were not as important to moose in our study area because of the milder winter conditions and the absence of wolf, as compared to the boreal region studied by Dussault et al. (2006). Additionally, these apparent discrepancies may be due to differences in methodology. Dussault et al. (2006) and Beyer et al. (2013) assessed moose habitat selection by comparing use to availability within individual home ranges (generally $>10 \text{ km}^2$ and more). Leclerc et al. (2012)

highlighted relationships between road density and moose counts in 60-km² aerial survey sampling plots. For this study, we evaluated availability within much smaller areas (3.14-km² landscape contexts) centered on plantations and naturally regenerated stands only. Moose may have reacted to these habitat variables at scales larger than what we measured (Rettie and Messier, 2000).

5. Management implications

The choice of a specific silvicultural treatment is driven by socio-economic and environmental values (Hartley, 2002). Our results suggest that the use of low-density plantations may represent an interesting compromise for competing local economies by favoring moose sport hunting in ITPA. Low-density plantations increased the availability of browsable stems, and moose reacted by browsing significantly more in low-density plantations compared to standard high-density plantations and naturally regenerated stands. In regions where intensive timber production is advocated, we predict that moose will select areas of low-density plantations over areas of high-density plantations. Over the long term, this strategy should help promote the success of moose hunters without compromising the profitability of the timber industry as we found that commercial trees were also larger on average in these stands. Higher food availability and cover provided by low-density plantations could also be favorable to other game species like white-tailed deer (Dumont et al., 2005) and snowshoe hare (*Lepus americanus*; Potvin et al., 2005).

Low-density plantations are neither the perfect nor the only available silvicultural treatment. The low-density plantation was designed to prevent the transition from mixed to conifer-

dominated stands in northern temperate mixed forests (Lafliche et al., 2000), and as such, it may not be as useful in pure coniferous or deciduous stands. Furthermore, we found that moose seemed to prefer low-density plantations for browsing, but their importance for other animal species remains to be elucidated. Low-density plantations should preferably be used in temperate regions where moose densities are high and where hunting success is a local priority.

We found high inter-site variations in field measurements both within and among silvicultural treatments. This result highlights the difficulty for forest managers to obtain precise stand characteristics by prescribing specific silvicultural treatments (St-Laurent et al., 2008). If moose are to be favored in ITPA, forest managers will obtain better results by aiming at obtaining specific post-harvest stand characteristics, i.e., high browsable stem availability and low vertical cover.

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Table A

Distribution of 108 sites and 540 sampling plots used to evaluate the effect of plantation type, pre-commercial thinning, and age class on proxies of habitat suitability for moose, use by moose, and forest productivity in Bas-Saint-Laurent, Québec, Canada.

Plantation type	Thinning treatment and age class					
	Thinned			Unthinned		
	5 years old	10 years old	15 years old	5 years old	10 years old	15 years old
Standard plantation (2000 stems/ha)	7 sites 35 plots	7 sites 35 plots	7 sites 35 plots	7 sites 35 plots	7 sites 35 plots	7 sites 35 plots
Experimental plantation (1350 stems/ha)	7 sites 35 plots	7 sites 35 plots	7 sites 35 plots	2 sites 10 plots	2 sites 10 plots	7 sites 35 plots
Naturally regenerated stands	– –	6 sites 30 plots	7 sites 35 plots	7 sites 35 plots	7 sites 35 plots	7 sites 35 plots

Table B

Food index and interspersions values for moose of stand types used to assess food availability and the density of edges between food and cover stands in landscape contexts surrounding spruce plantations and naturally regenerated stands in Bas-Saint-Laurent, Québec, Canada. Food and interspersions values are based on Dussault et al. (2006). Availability (%) of each stand type within the study area is also shown.

Stand type	Description	Food index value	Interspersion value	Availability (%)
Mi10	Mixed or deciduous regenerating stand ^a ; <10 years old	1.00	Food	18.1
Mi30	Mixed with shade-intolerant deciduous trees ^b ; 30 years old	0.50	–	7.0
Mi50	Mixed with shade-intolerant deciduous trees ^b ; ≥50 years old	0.40	Cover	8.6
Mt50	Mixed with shade-tolerant deciduous trees ^c ; ≥50 years old	1.00	Food	11.1
Di50	Deciduous with shade-intolerant trees ^c ; ≥50 years old	0.50	–	2.3
Dt50	Deciduous with shade-tolerant trees ^c ; ≥50 years old	1.00	Food	6.2
C10	Coniferous regenerating stand; <10 years old	0.30	–	11.6
CF30	Coniferous with balsam fir; ≥30 years old	0.15	Cover	24.4
CS30	Coniferous with black spruce; ≥30 years old	0.05	Cover	5.8
UNP	Unproductive area ^d	0.10	–	1.3
OTHER	Other weakly represented forested or non-forested area ^e	0.00	–	3.6

^a Including natural disturbances, like windthrows and insect epidemics.

^b Mostly *Betula papyrifera*, *Populus sp.*, and *Prunus sp.*

^c Mostly *B. alleghaniensis* and *Acer sp.*

^d Including bogs, fens, and alder stands.

^e Including water bodies.

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Appendix A

See Table A.

Appendix B

See Table B.

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